Sound as Sinusoids

Intro to Audio Tech 1
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Wednesday, August 20

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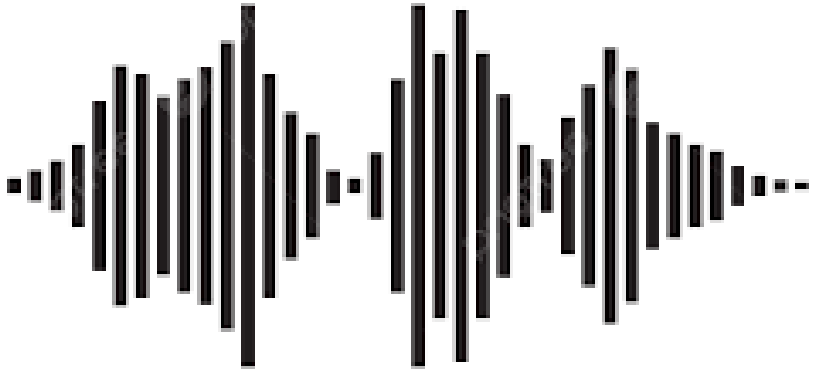
Lesson Outline

- How sound travels and perception
- Properties of Sinusoids
- Signal Flow
- Max Demo
- Up Next…



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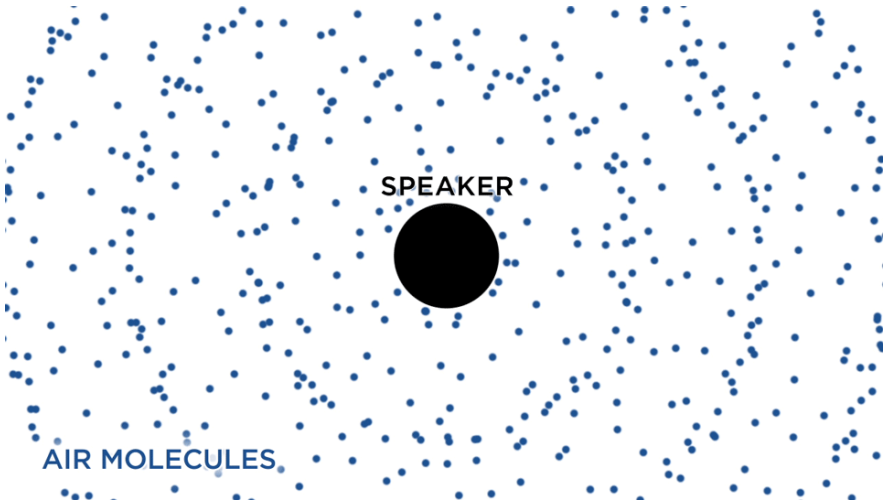
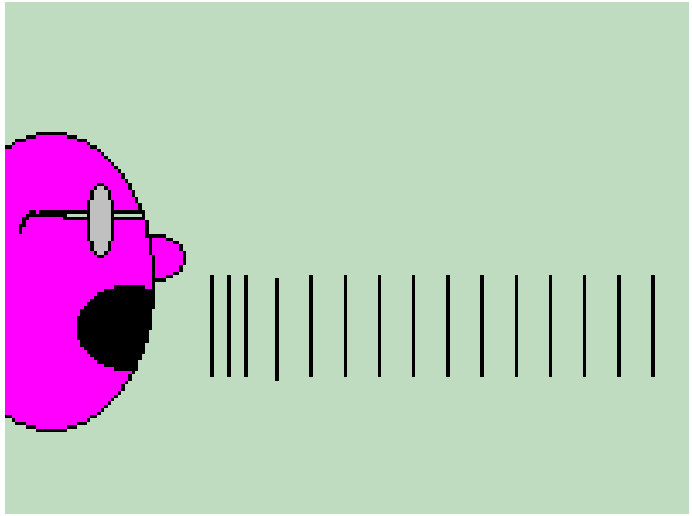
Sound Waves

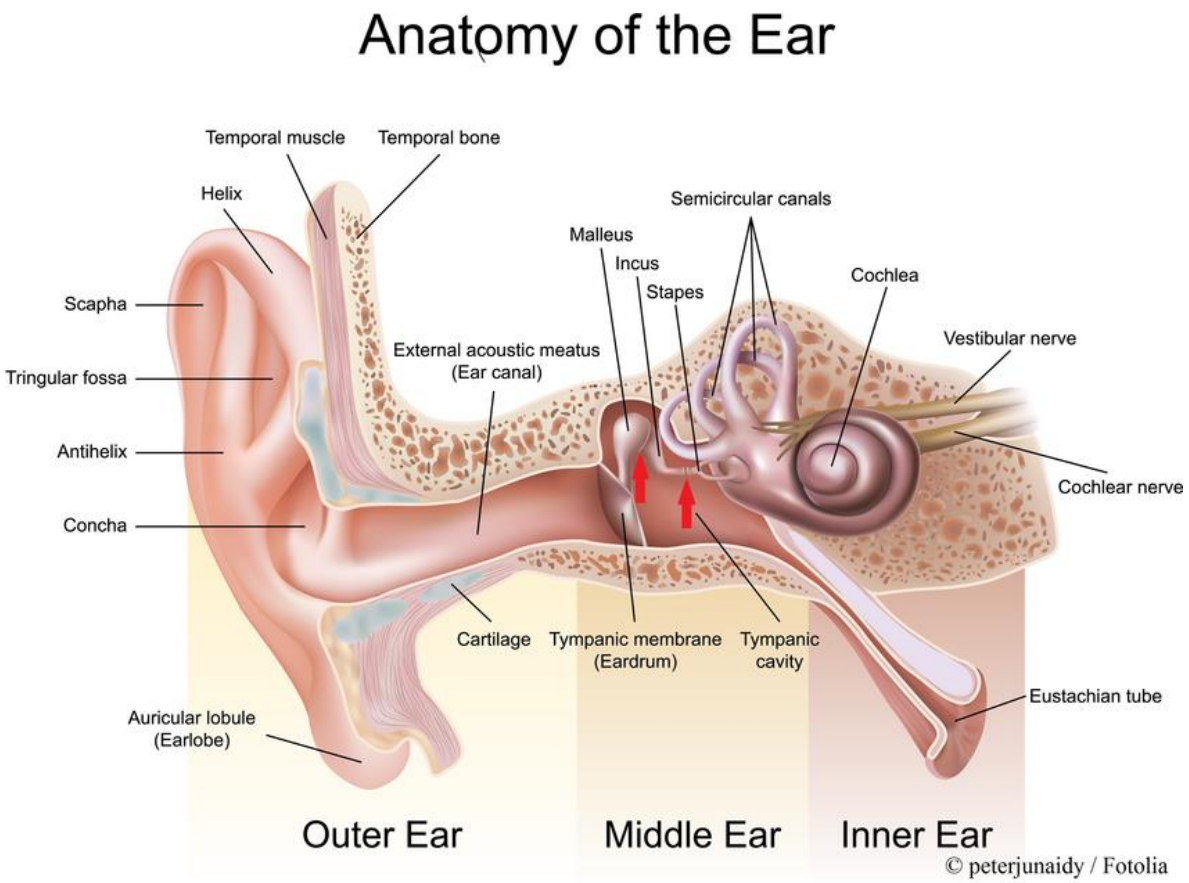
- Air pressure changes that vibrate our eardrums (or microphones)
  - “High pressure front” from the impulse
- Longitudinal Wave
  - Disturbance
  - Restorative force
  - Compression and Rarefaction
- Sound travels at ~343 m/s





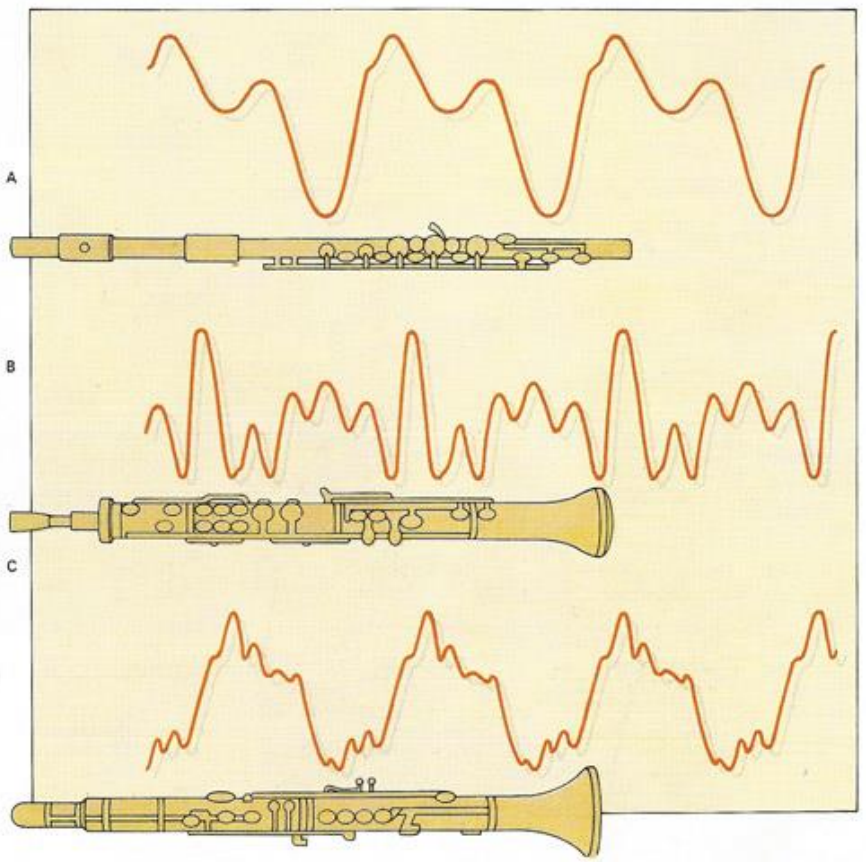
Impulse to Perception

- Sound source: clap
- Disturbs air molecules that vibrate the eardrum
- Eardrum moves the ossicles
- Ossicles amplify the “sound” into the cochlea
  - (through the oval window and Eustachian tube)
- Fluid in the cochlea vibrates and moves tiny “hairs” (nerve endings)
- Nerve endings transform the vibrations into electrical impulses
- Electrical impulses go to the auditory nerve into the temporal lobe for perception

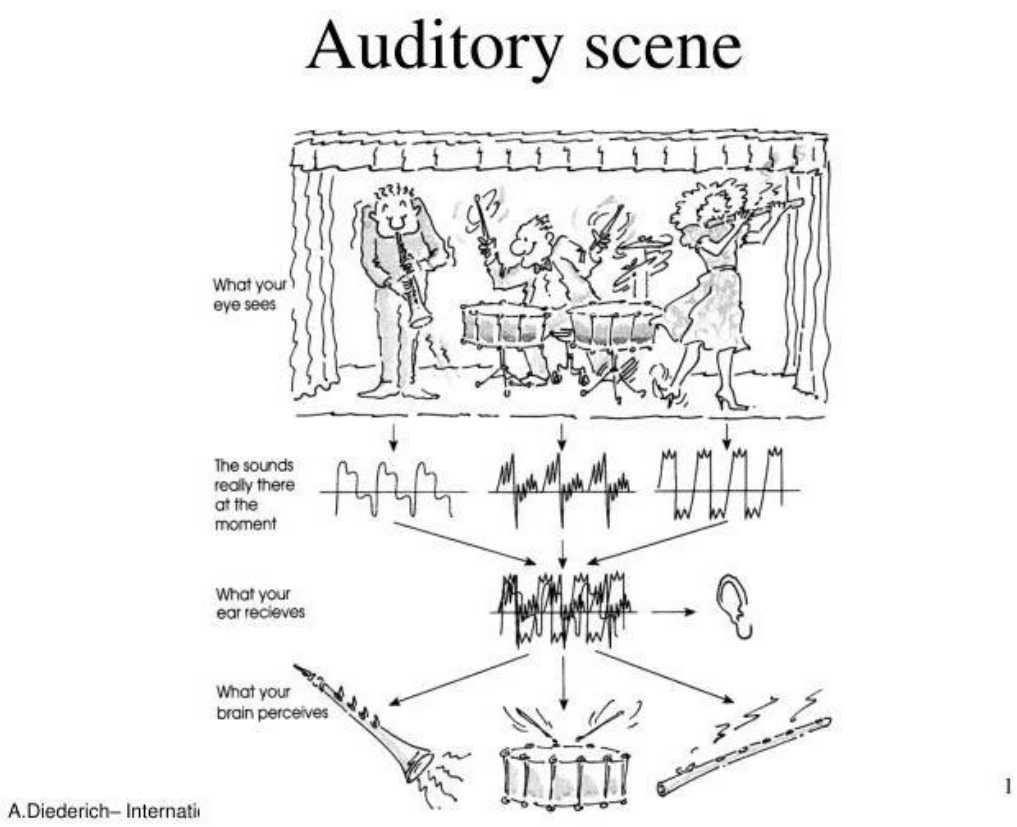


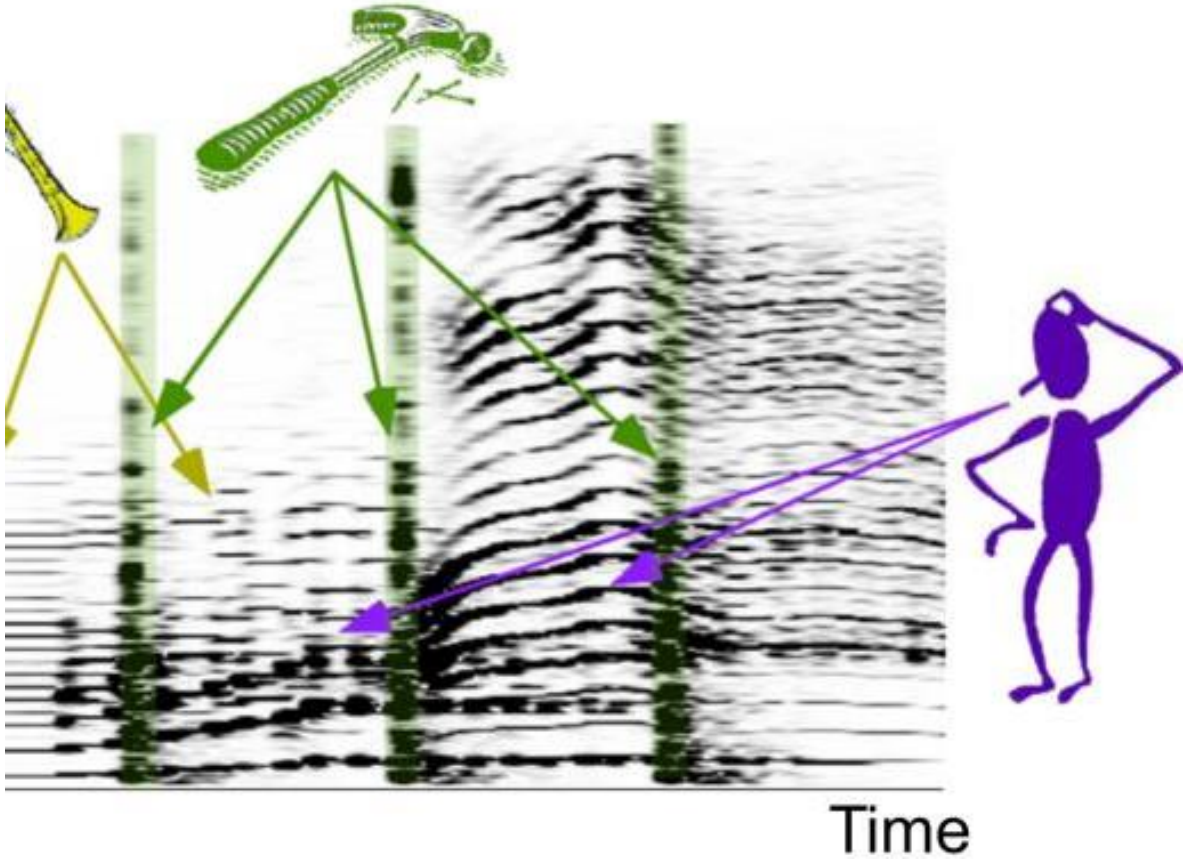
Parameters of sound

- We are used to talking about sound through 3 parameters
  - Frequency – how we distinguish higher or lower pitch
  - Amplitude – how we distinguish loud and soft sounds
  - Waveform – how we distinguish different timbres



Auditory Scene Analysis



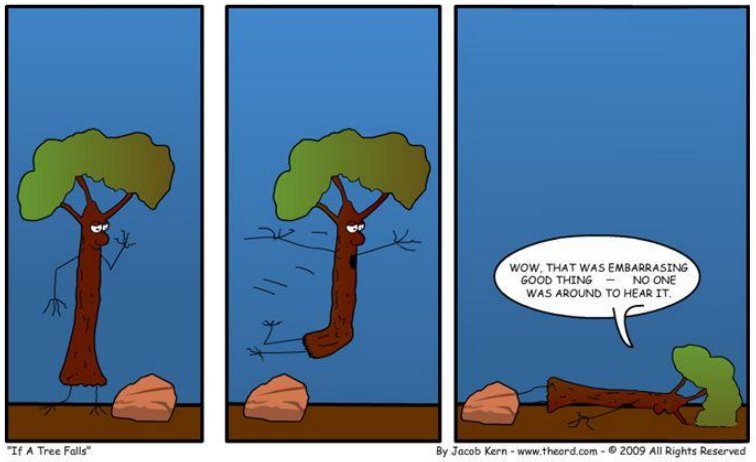


Let’s Clarify Some Terms

- Sensation vs Perception
- We can liken to…

Characteristic	Parameter	Perception
Amplitude	Intensity/Loudness	Louds and softs
Frequency	Pitch	Highs and lows
Waveform (Spectra and Envelope)	Timbre	Sound color

Sensation and Perception

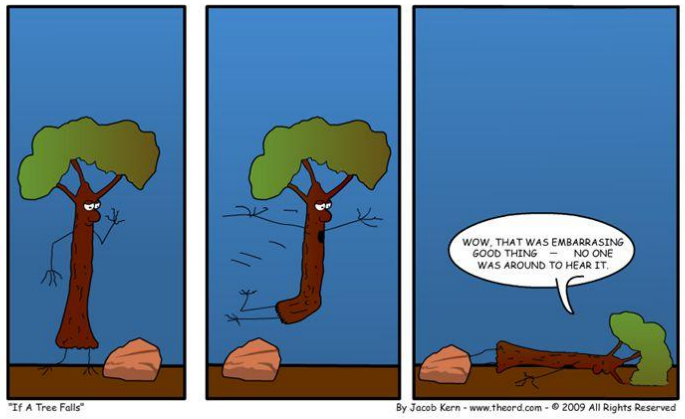


If a tree falls in the woods and no one is around to hear it, does it make a sound?

Sensation and Perception

- Sensation is the input obtained by our sensory receptors
  - Our ability to detect sensory input
  - Nerves in cochlea are stimulated based on the frequency/amplitude of the sound input
- Perception is the processing and interpretation of that input
  - Our brain deciding what the sensation means
  - Temporal lobe process and decides the sound is a musical pitch

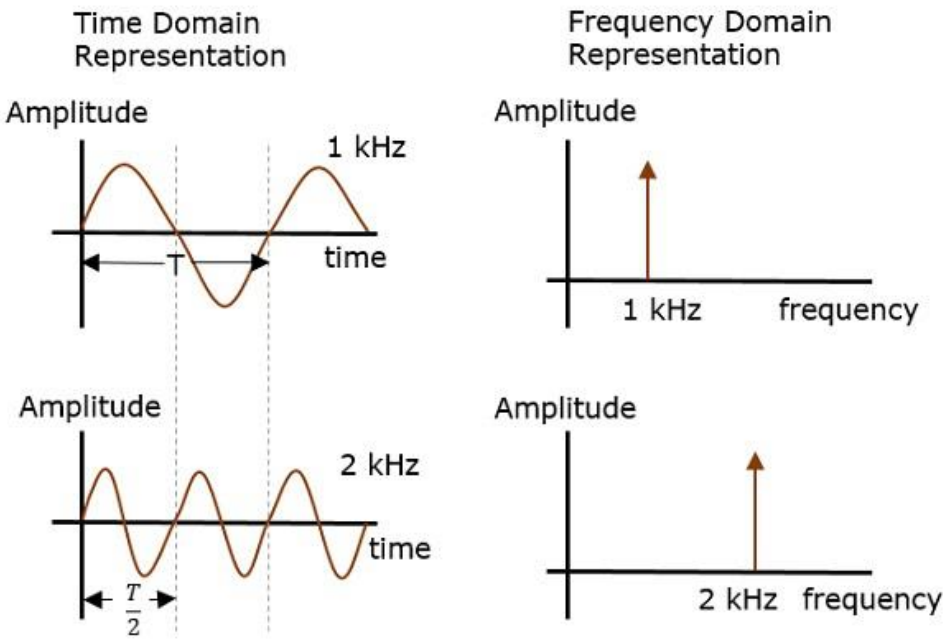
Sensation and Perception



If a tree falls in the woods and no one is around to hear it, does it make a sound?

How do we view sound?

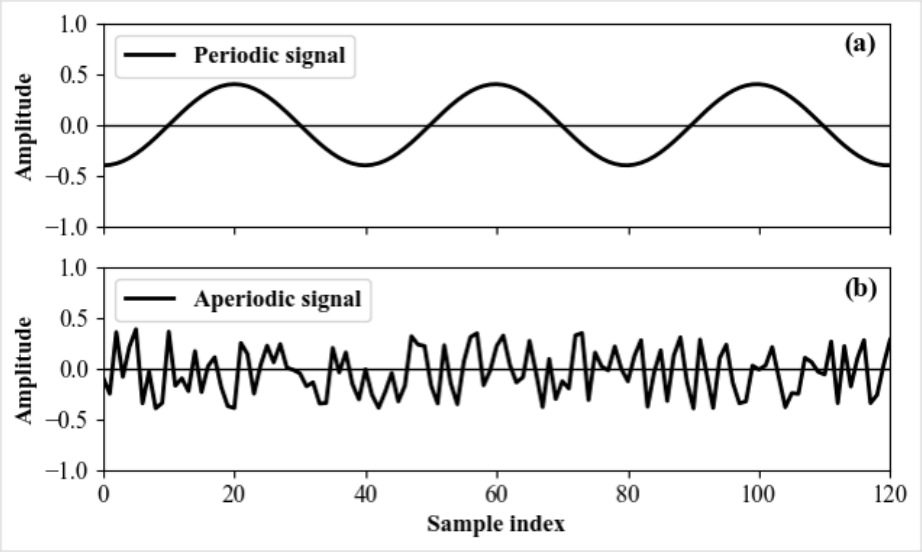
- Time domain
  - Waveform
  - Tells you information about amplitude over time
- Frequency domain
  - Magnitude Spectrum
  - Tells you information about frequencies and their strength at a given time



Sinusoids and Periodic Sound

Periodic vs Aperiodic

- If a sound wave is periodic (follows a repeated pattern), we can describe it using:
  - Amplitude – How big are the peaks of the wave
  - Frequency – How fast does the wave cycle through peaks
  - Phase – Where in the cycle are we
- If a sound doesn't follow a pattern, it is aperiodic or noise

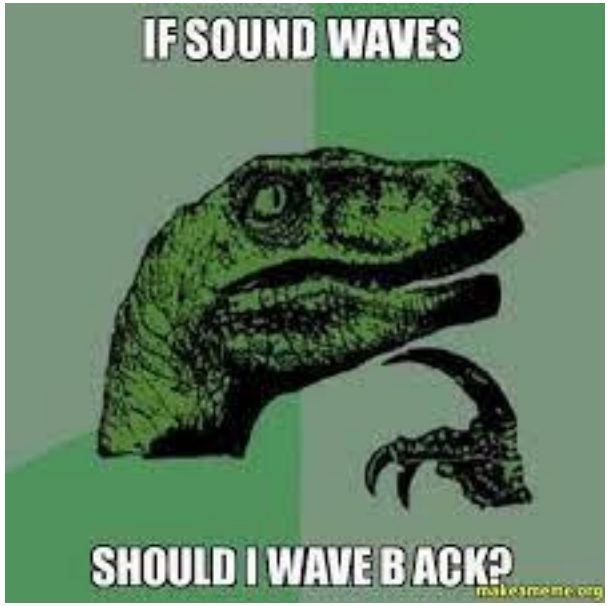


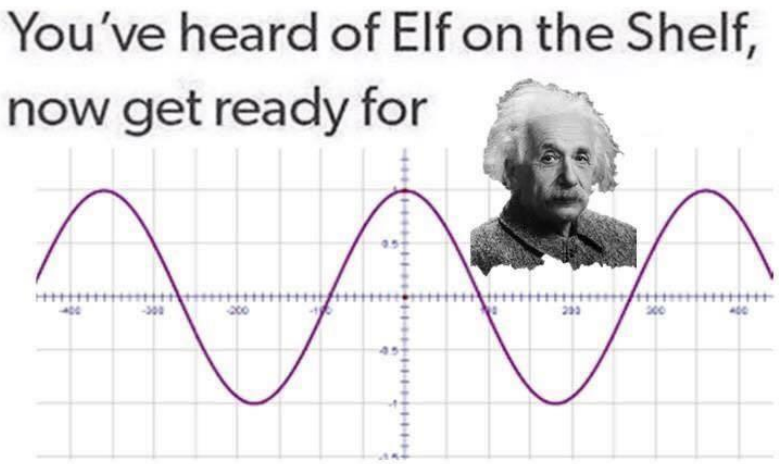
Sinusoids

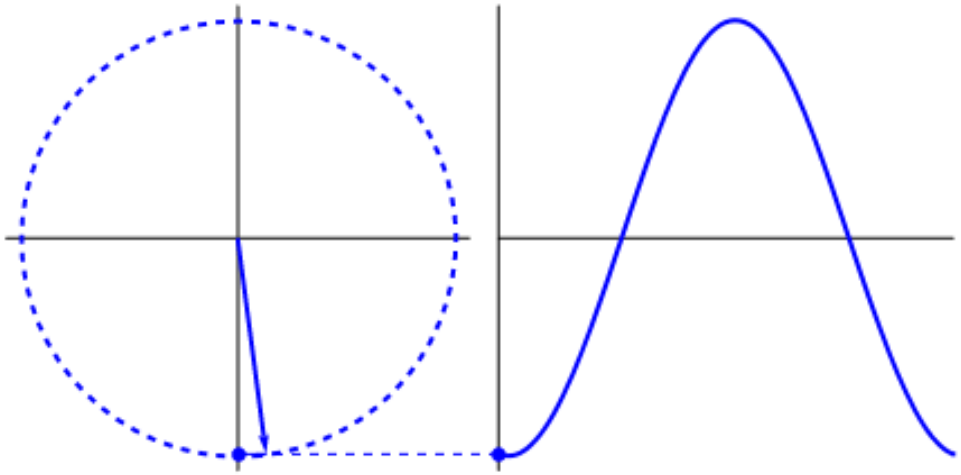
- A sinusoid is the simplest waveform
- Has only 1 frequency
- Sine waves and cosine waves are sonically indistinguishable but characterized by a phase shift (an important difference later)
- Reminders
  - One pattern repetition is called a cycle
  - The number of cycles in a second is the fundamental frequency

Sinusoids

$$x[n] = a\cos(\omega n + \phi)$$

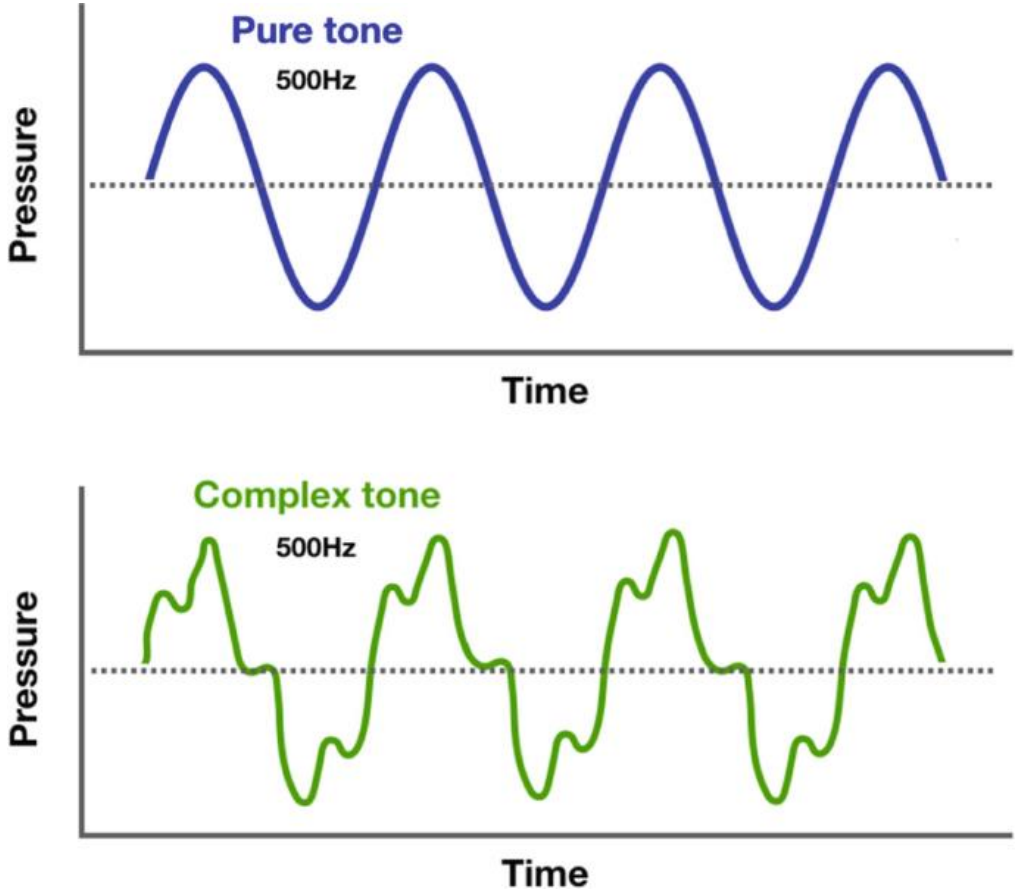






Sinusoids and Complex Tones

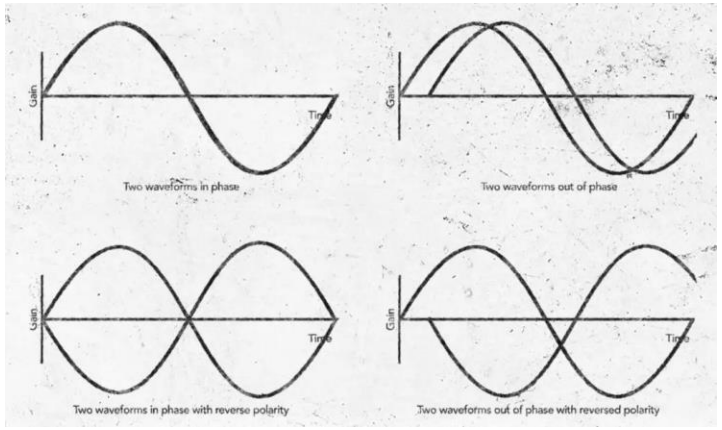
- Sine/cosine waves have a single frequency
- In the real world, almost all sounds are made up of multiple frequencies
- The sum of all those harmonically related frequencies still results in a periodic wave having the fundamental frequency
- More on this later…

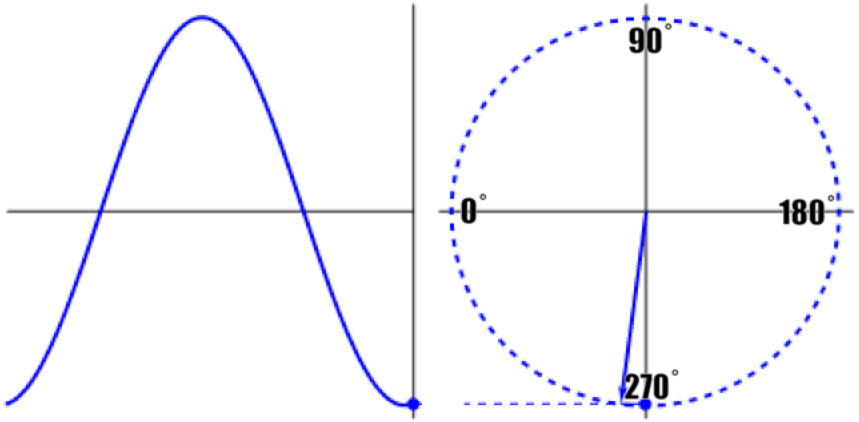


Phase

What is phase?

- The “starting point” of a periodic waveform on the y axis
- In phase or phase aligned
  - when two signals ”start” at the same time
- Out of phase
  - When a signal is delayed with respect to the other signal
- Reverse polarity or phase-inverted
  - When a signal is perfectly out of phase with another





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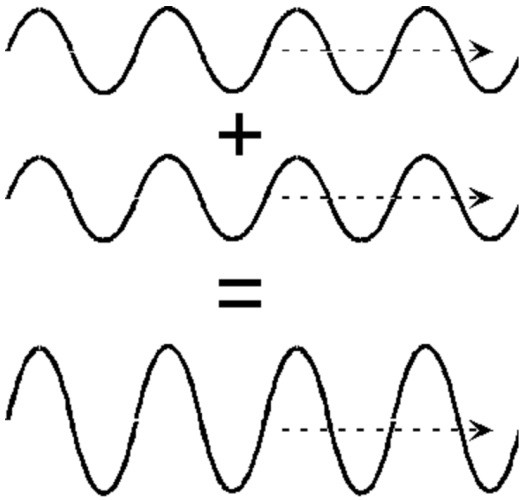
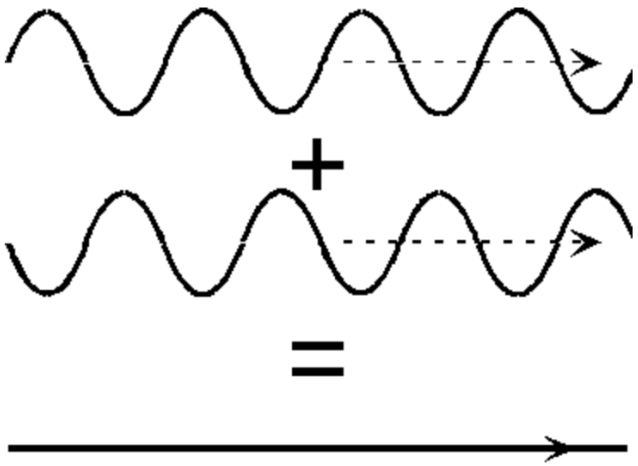
Phase

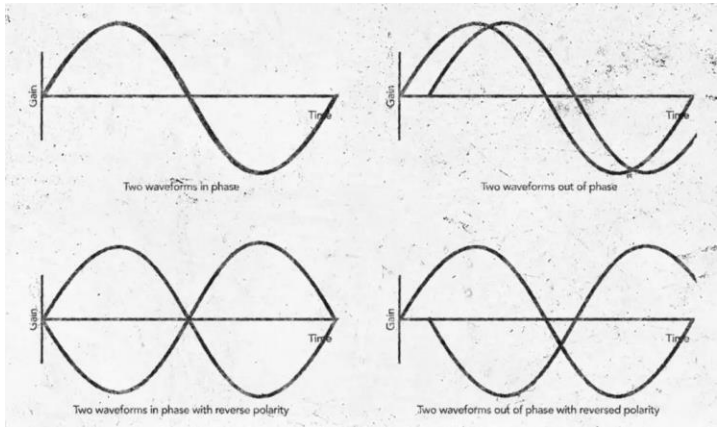
- Constructive and Destructive Interference
- May not be distinguishable alone but very important in systems

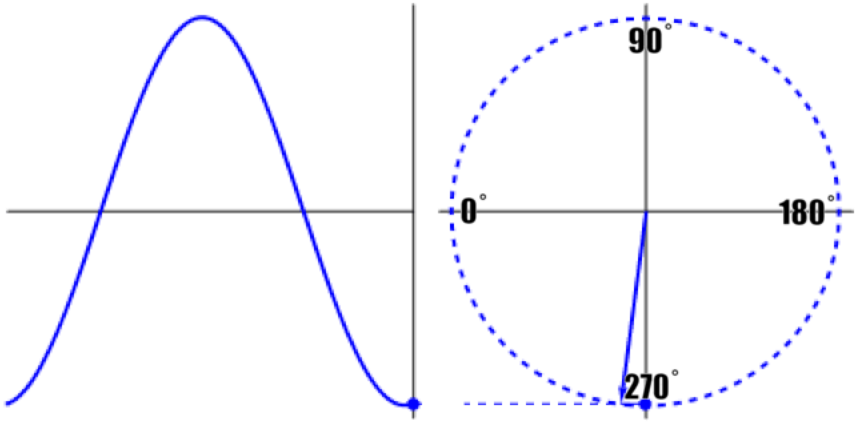




Why is phase important?

- In controlled settings, humans may be able to detect polarity
- Every filter is built off phase shifts
  - Frequency-dependent cancellation
  - Phase reinforcement
  - time-varying phase shifts lead to phasing or flanging
- Phase shifts distort musical signals during recording or playback
  - Phase distortion

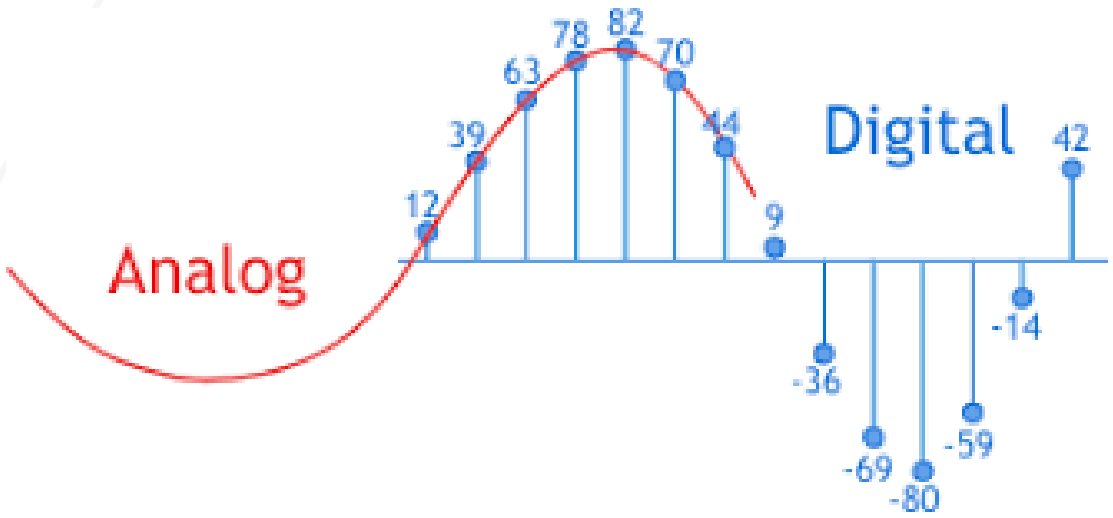


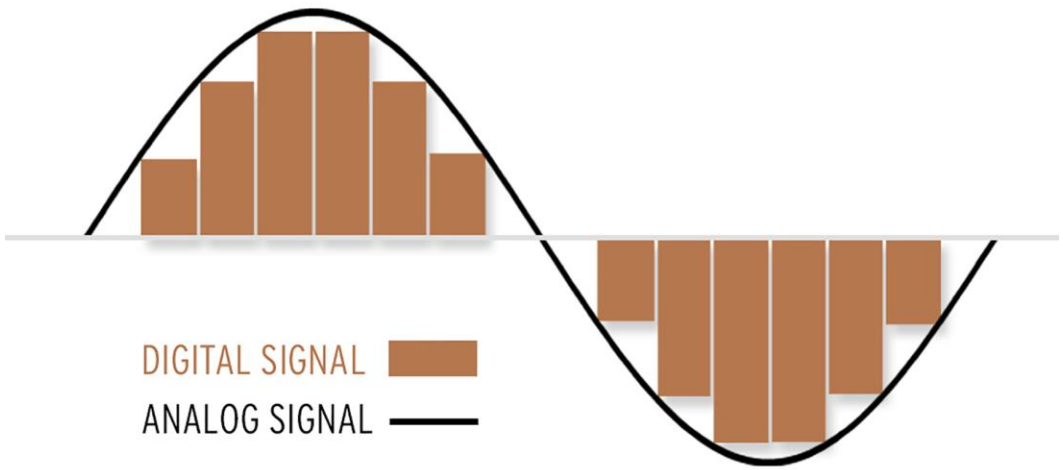


Aside on Signal Flow and Digital Audio

Analog to Digital

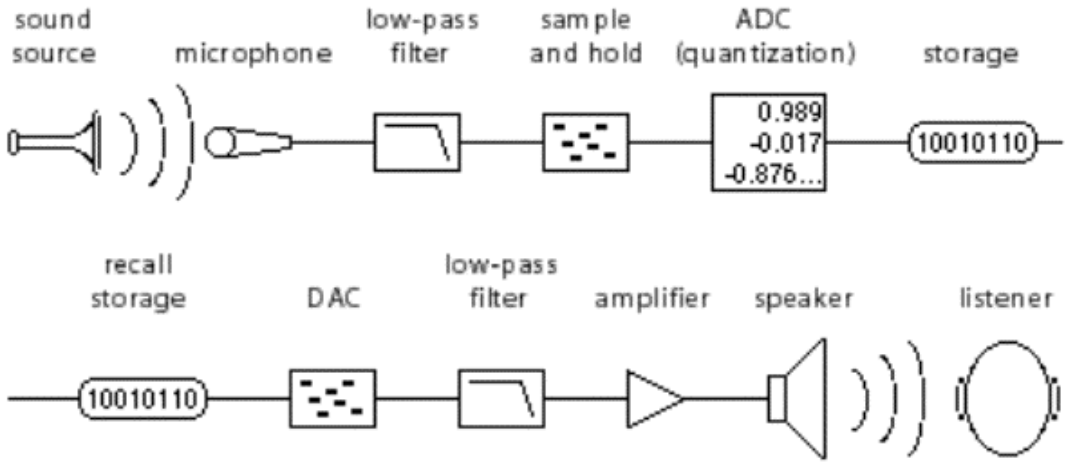
- Analog -> Continuous
- Digital -> Discrete





Analog to Digital

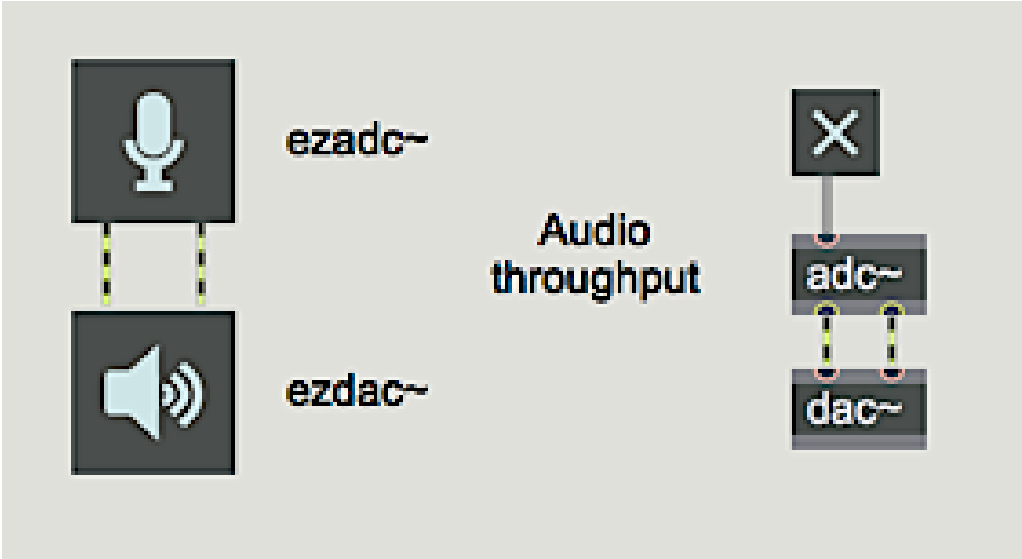
- Analog signals becoming digital through sampling and quantization
- Analog to Digital Converter (ADC)
  - convert voltages into binary numbers at some rate
- Digital to Analog Converter (DAC)
  - Convert binary numbers into voltages at some rate (with smoothing and amplification)



Sinusoids in Max

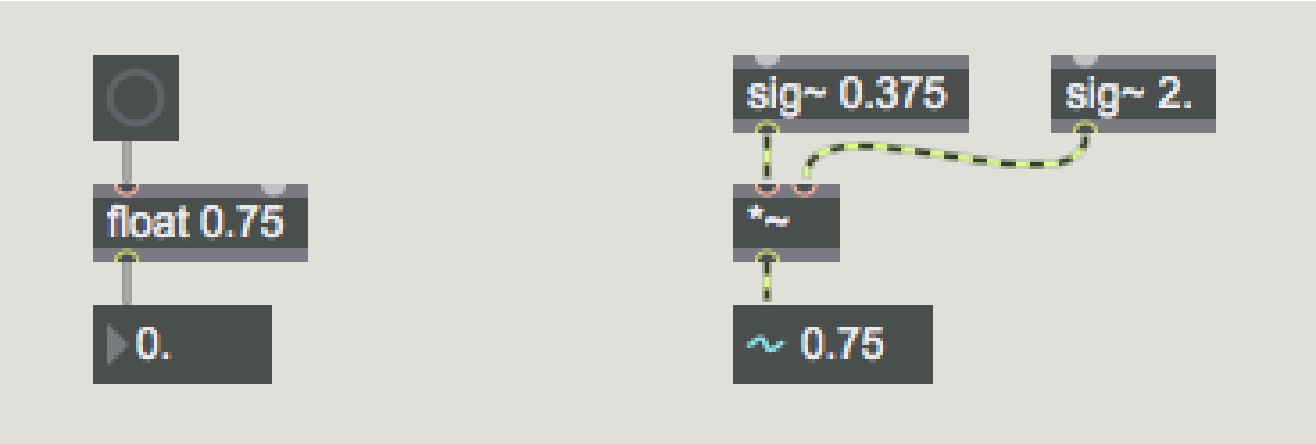
Getting sound in and out

- ADC~ and DAC~
- Analog to digital converter
- Digital to analog converter
- Convert digital discrete stream to continuous analog



Using signals

- ~ is added to the end of objects to interface with MSP
- Signals are sent continuously (without your intervention)
- Max schedules at the ms, but will match to sampling rate



Up Next…

- Readings!
- Play with Max
  - Email helpdesk@design.gatech.edu if you are having trouble
- Office hours and exercises